

**CSE422: Artificial intelligence**

# **Naïve-Bayes Classifier**

Asif Shahriar  
Lecturer, CSE, BRACU

# BAYES' Rule

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- From the formula of conditional probability  $P(x | y) = P(x, y) / P(y)$ , we can infer:  
 $P(x, y) = P(x | y) \times P(y)$  [called the product rule]

- Similarly,  $P(x, y) = P(y | x) \times P(x)$

- Dividing we get:

$$P(x|y) = \frac{P(y|x)P(x)}{P(y)}$$

- In other words:  $P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$

- Why is this SO important?
  - It is hard to know the cause, but it is easy to see the effect
  - Using this formula, we can infer the cause by analyzing the effect

# BAYES' Rule

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$$\boxed{P(x|y)} = \frac{\boxed{P(y|x)}}{\boxed{P(y)}} \boxed{P(x)}$$

Posterior probability

Likelihood

Predictor-prior probability

Class-prior probability

# BAYES' Rule

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| Temperature (F) | Play Tennis |
|-----------------|-------------|
| 70              | Yes         |
| 32              | No          |
| 65              | No          |
| 75              | Yes         |
| 30              | No          |
| 75              | Yes         |
| 72              | No          |

- Consider two boolean RVs, A: “We will play tennis” and B: “Warm day (Temp  $\geq$  50 F)”
  - Think ML: Temperature is “**feature**”, Play tennis is “**target**”
- If it is a warm day, what is the probability that we will play tennis?
  - Think ML: Given a **feature**, predict the **target**

# BAYES' Rule

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|-----------------|-------------|
| 70              | Yes         |
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| 65              | No          |
| 75              | Yes         |
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| 75              | Yes         |
| 72              | No          |

- If it is a warm day, what is the probability that we will play tennis?
- $P(b | a) = 1$
- $P(a) = 3/7$
- $P(b) = 5/7$
- Therefore,  $P(a | b) = 0.6$ 
  - Think ML: **Output = YES**

# Why BAYES is DIFFICULT in Practice

| Wind | Humidity | Temperature | Play Tennis |
|------|----------|-------------|-------------|
| 5    | 95       | 70          | Yes         |
| 10   | 80       | 32          | No          |
| 20   | 80       | 65          | No          |
| 10   | 85       | 75          | Yes         |
| 8    | 35       | 30          | No          |
| 8    | 35       | 75          | Yes         |
| 25   | 35       | 72          | No          |

- Now let there are FOUR boolean RVs: A (“Play tennis”), B(“Warm day”), C(“Dry day”), D(“Windy day”) [Think ML: Features: B, C, D, Target: A]
- We want to find the probability of playing tennis in a warm, dry, windy day  $P(a | b, c, d)$
- For this we need the likelihood  $P(b, c, d | a)$
- Computing this is REALLY difficult and makes it hard to use BAYES in real applications
  - For this problem it is actually easy (do it)
  - But this is a small table with 7 entries only
  - In real world, to compute such a likelihood with good accuracy would require an extensive record with millions of entries – very expensive and difficult

# Naïve-Bayes Classifier

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- Computing the likelihood  $P(b, c, d | a)$  is difficult in real world applications
- However, we can APPROXIMATE the calculation, making it practical for real use
- The **Naïve assumption**: All **features** are **conditionally independent** of each other given the target label
- In other words, B, C, D are conditionally independent given A
- So, using our formula for conditional independence, we get:
- $P(b, c, d | a) = P(b | a) \times P(c | a) \times P(d | a)$
- This is much easier to calculate from records!
- The Naïve Bayes classifier often does surprisingly well, outperforming more sophisticated classification methods
- $P(X_1, X_2, \dots, X_n | C) = P(X_1 | C)P(X_2 | C) \dots P(X_n | C)$

# Approximation using Naïve-BAYES

| Wind | Humidity | Temperature | Play Tennis |
|------|----------|-------------|-------------|
| 5    | 95       | 70          | Yes         |
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| 20   | 80       | 65          | No          |
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| 8    | 35       | 75          | Yes         |
| 25   | 35       | 72          | No          |

- We want to find the probability of playing tennis in a warm, dry, windy day  $P(a | b, c, d)$
- $P(a | b, c, d) = P(b, c, d | a) \times P(a) / P(b, c, d)$
- Do it yourself!

Based on the Naïve Bayes approximation using the provided data:

$P(\text{Yes} | \text{warm, dry, windy}) \propto 0$  (since no windy "Yes" example exists).

$P(\text{No} | \text{warm, dry, windy}) \propto 0.0357$ .

Therefore, the classifier predicts "No" (do not play tennis)



# Naïve-BAYES Example

*PlayTennis: training examples*

| Day | Outlook  | Temperature | Humidity | Wind   | PlayTennis |
|-----|----------|-------------|----------|--------|------------|
| D1  | Sunny    | Hot         | High     | Weak   | No         |
| D2  | Sunny    | Hot         | High     | Strong | No         |
| D3  | Overcast | Hot         | High     | Weak   | Yes        |
| D4  | Rain     | Mild        | High     | Weak   | Yes        |
| D5  | Rain     | Cool        | Normal   | Weak   | Yes        |
| D6  | Rain     | Cool        | Normal   | Strong | No         |
| D7  | Overcast | Cool        | Normal   | Strong | Yes        |
| D8  | Sunny    | Mild        | High     | Weak   | No         |
| D9  | Sunny    | Cool        | Normal   | Weak   | Yes        |
| D10 | Rain     | Mild        | Normal   | Weak   | Yes        |
| D11 | Sunny    | Mild        | Normal   | Strong | Yes        |
| D12 | Overcast | Mild        | High     | Strong | Yes        |
| D13 | Overcast | Hot         | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild        | High     | Strong | No         |

Step 1 – **Learning**: From the training data, we will **LEARN the conditional probability** distribution of each **Feature** (Outlook, Temperature, Humidity, Wind) Given the **Target** (PlayTennis)

# Naïve-BAYES Example

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| D8  | Sunny    | Mild        | High     | Weak   | No         |
| D9  | Sunny    | Cool        | Normal   | Weak   | Yes        |
| D10 | Rain     | Mild        | Normal   | Weak   | Yes        |
| D11 | Sunny    | Mild        | Normal   | Strong | Yes        |
| D12 | Overcast | Mild        | High     | Strong | Yes        |
| D13 | Overcast | Hot         | Normal   | Weak   | Yes        |
| D14 | Rain     | Mild        | High     | Strong | No         |

Learning Phase

| Outlook  | Play=Yes | Play=No |
|----------|----------|---------|
| Sunny    | 2/9      | 3/5     |
| Overcast | 4/9      | 0/5     |
| Rain     | 3/9      | 2/5     |

| Temperature | Play=Yes | Play=No |
|-------------|----------|---------|
| Hot         | 2/9      | 2/5     |
| Mild        | 4/9      | 2/5     |
| Cool        | 3/9      | 1/5     |

| Humidity | Play=Yes | Play=No |
|----------|----------|---------|
| High     | 3/9      | 4/5     |
| Normal   | 6/9      | 1/5     |

| Wind   | Play=Yes | Play=No |
|--------|----------|---------|
| Strong | 3/9      | 3/5     |
| Weak   | 6/9      | 2/5     |

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Play}=\text{No}) = 5/14$$

Caution: This is a **CONDITIONAL** probability table, NOT Joint Probability Distribution Table like toothache, cavity tables

# Naïve-BAYES Example

Step 2 – **Testing**: Using what we learned (likelihoods) during training, we will **PREDICT** the label (target) for a new, unseen input data

## Learning Phase

| Outlook  | Play=Yes | Play=No |
|----------|----------|---------|
| Sunny    | 2/9      | 3/5     |
| Overcast | 4/9      | 0/5     |
| Rain     | 3/9      | 2/5     |

| Humidity | Play=Yes | Play=No |
|----------|----------|---------|
| High     | 3/9      | 4/5     |
| Normal   | 6/9      | 1/5     |

$$P(\text{Play}=\text{Yes}) = 9/14$$

| Temperature | Play=Yes | Play=No |
|-------------|----------|---------|
| Hot         | 2/9      | 2/5     |
| Mild        | 4/9      | 2/5     |
| Cool        | 3/9      | 1/5     |

| Wind   | Play=Yes | Play=No |
|--------|----------|---------|
| Strong | 3/9      | 3/5     |
| Weak   | 6/9      | 2/5     |

$$P(\text{Play}=\text{No}) = 5/14$$

## Test Phase

- Given a new instance, predict its label

$\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

- Look up tables achieved in the learning phase

$$\begin{aligned} P(\text{Outlook}=\text{Sunny} | \text{Play}=\text{Yes}) &= 2/9 & P(\text{Outlook}=\text{Sunny} | \text{Play}=\text{No}) &= 3/5 \\ P(\text{Temperature}=\text{Cool} | \text{Play}=\text{Yes}) &= 3/9 & P(\text{Temperature}=\text{Cool} | \text{Play}=\text{No}) &= 1/5 \\ P(\text{Humidity}=\text{High} | \text{Play}=\text{Yes}) &= 3/9 & P(\text{Humidity}=\text{High} | \text{Play}=\text{No}) &= 4/5 \\ P(\text{Wind}=\text{Strong} | \text{Play}=\text{Yes}) &= 3/9 & P(\text{Wind}=\text{Strong} | \text{Play}=\text{No}) &= 3/5 \\ P(\text{Play}=\text{Yes}) &= 9/14 & P(\text{Play}=\text{No}) &= 5/14 \end{aligned}$$

- Decision making

$$P(\text{Yes} | \mathbf{x}') \approx [P(\text{Sunny} | \text{Yes})P(\text{Cool} | \text{Yes})P(\text{High} | \text{Yes})P(\text{Strong} | \text{Yes})]P(\text{Play}=\text{Yes}) = 0.0053$$

$$P(\text{No} | \mathbf{x}') \approx [P(\text{Sunny} | \text{No})P(\text{Cool} | \text{No})P(\text{High} | \text{No})P(\text{Strong} | \text{No})]P(\text{Play}=\text{No}) = 0.0206$$

Given the fact  $P(\text{Yes} | \mathbf{x}') < P(\text{No} | \mathbf{x}')$ , we label  $\mathbf{x}'$  to be "No".

# Naïve-BAYES for Continuous-valued Features

- Features are not always discrete, sometimes they are continuous
- Example: Temperature is a naturally continuous RV
  - Yes: 25.2, 19.3, 18.5, 21.7, 20.1, 24.3, 22.8, 23.1, 19.8
  - No: 27.3, 30.1, 17.4, 29.5, 15.1
- Problem: Very unlikely that we will find an exact value of temperature
  - E.g.  $P(\text{Play} = \text{YES} \mid \text{Temp} = 25.8)$ , but  $\text{Temp} = 25.8$  DOES NOT EXIST in data
- Solution: Model continuous probabilities with **Normal Distribution**

$$\hat{P}(X_j \mid C = c_i) = \frac{1}{\sqrt{2\pi}\sigma_{ji}} \exp\left(-\frac{(X_j - \mu_{ji})^2}{2\sigma_{ji}^2}\right)$$

$\mu_{ji}$  : mean (average) of feature values  $X_j$  of examples for which  $C = c_i$

$\sigma_{ji}$  : standard deviation of feature values  $X_j$  of examples for which  $C = c_i$

# Naïve-BAYES for Continuous-valued Features

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$$\hat{P}(X_j | C = c_i) = \frac{1}{\sqrt{2\pi}\sigma_{ji}} \exp\left(-\frac{(X_j - \mu_{ji})^2}{2\sigma_{ji}^2}\right)$$

$\mu_{ji}$  : mean (average) of feature values  $X_j$  of examples for which  $C = c_i$

$\sigma_{ji}$  : standard deviation of feature values  $X_j$  of examples for which  $C = c_i$

**Learning Phase:** for  $\mathbf{X} = (X_1, \dots, X_n)$ ,  $C = c_1, \dots, c_L$

**Output:**  $n \times L$  normal distributions and  $P(C = c_i)$   $i = 1, \dots, L$

**Test Phase:** Given an unknown instance  $\mathbf{X}' = (a'_1, \dots, a'_n)$

- Instead of looking-up tables, calculate conditional probabilities with all the normal distributions achieved in the learning phase

# Naïve-BAYES for Continuous-valued Features

- Let's consider the temperature example again
  - Yes: 25.2, 19.3, 18.5, 21.7, 20.1, 24.3, 22.8, 23.1, 19.8
  - No: 27.3, 30.1, 17.4, 29.5, 15.1
- Estimate mean and variance for each class

$$\mu = \frac{1}{N} \sum_{n=1}^N x_n, \quad \sigma^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2$$

$$\mu_{Yes} = 21.64, \quad \sigma_{Yes} = 2.35$$

$$\mu_{No} = 23.88, \quad \sigma_{No} = 7.09$$

- **Learning Phase:** output two Gaussian models for  $P(\text{temp}|\text{C})$

$$\hat{P}(x | Yes) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x - 21.64)^2}{2 \times 2.35^2}\right) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x - 21.64)^2}{11.09}\right)$$

$$\hat{P}(x | No) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x - 23.88)^2}{2 \times 7.09^2}\right) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x - 23.88)^2}{50.25}\right)$$



# Relevant Issues to consider for Naïve-BAYES

- **Violation of Independence Assumption**

- For most real worlds tasks,  $P(X_1, \dots, X_n | C) \neq P(X_1 | C) \dots P(X_n | C)$  [i.e. features are not independent]
- Nonetheless, Naïve-BAYES works surprisingly well anyway
- Check this out for a comparison against other models: [notebook](#)

- **Zero conditional probability Problem**

- If no example contains the feature value  $X_j = a_{jk}$ ,  $\hat{P}(X_j = a_{jk} | C = c_i) = 0$
- In this circumstance,  $\hat{P}(x_1 | c_i) \dots \hat{P}(a_{jk} | c_i) \dots \hat{P}(x_n | c_i) = 0$  during test
- For a remedy, conditional probabilities re-estimated with

$$\hat{P}(X_j = a_{jk} | C = c_i) = \frac{n_c + mp}{n + m}$$

$n_c$  : number of training examples for which  $X_j = a_{jk}$  and  $C = c_i$

$n$  : number of training examples for which  $C = c_i$

$p$  : prior estimate (usually,  $p = 1/t$  for  $t$  possible values of  $X_j$ )

$m$  : weight to prior (number of "virtual" examples,  $m \geq 1$ )

# Continuous Naïve-BAYES Practice

| Temperature | Play |
|-------------|------|
| 25.2        | Yes  |
| 27.3        | No   |
| 19.3        | Yes  |
| 18.5        | Yes  |
| 30.1        | No   |
| 21.7        | Yes  |
| 17.4        | No   |

| Temperature | Play |
|-------------|------|
| 20.1        | Yes  |
| 24.3        | Yes  |
| 29.5        | No   |
| 22.8        | Yes  |
| 23.1        | Yes  |
| 19.8        | Yes  |
| 15.1        | No   |

Determine whether we will play or not on a day with temperature = 25.8, using continuous-valued Naive-Bayes.

$$\mu_{Yes} = 21.64, \sigma_{Yes} = 2.35$$

$$\mu_{No} = 23.88, \sigma_{No} = 7.09$$